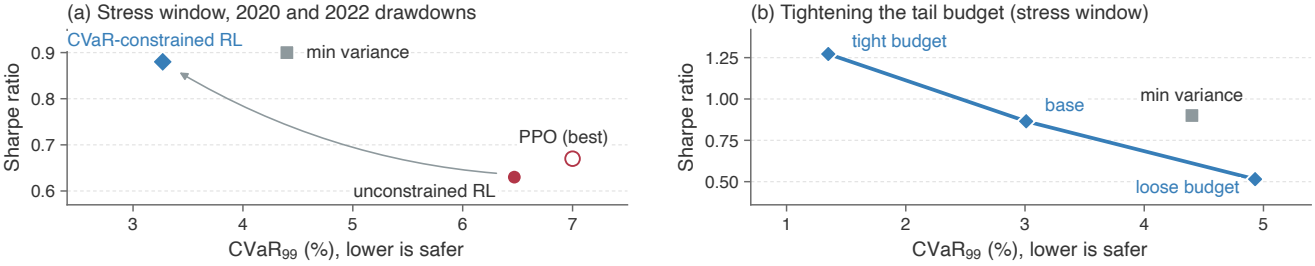
(a) Stress window, 2020 and 2022 drawdowns
CVaR-constrained RL
min variance
PPO (best)
unconstrained RL
Sharpe ratio
0.9
0.8
0.7
0.6
3
4
5
6
7
CVaR99 (%), lower is safer
(b) Tightening the tail budget (stress window)
tight budget
base
min variance
loose budget
Sharpe ratio
1.25
1.00
0.75
0.50
1
2
3
4
5
CVaR99 (%), lower is safer